Journal of Environmental Chemical Engineering 11 (2023) 110163

Contents lists available at ScienceDirect

Journal of Environmental Chemical Engineering

journal homepage: www.elsevier.com/locate/jece

Kinetic study on the effects of elevated temperature and micro aeration in volatile fatty acids production by anaerobic fermentation of alkaline-pretreated oil palm empty fruit bunch

Andri Sanjaya [a,b], Deni Frans Sakka [c], Rara Ayu Lestary [b], Roberty Wisnu Alvania Kartika [c], Desi Riana Saputri [a], Damayanti Damayanti [a], Ria Millati [d], Wiratni Budhijanto [b,e,*]

[a] Chemical Engineering Department, Institut Teknologi Sumatera, Jalan Terusan Ryacudu, Lampung Selatan 35365, Indonesia
[b] Chemical Engineering Department, Universitas Gadjah Mada, Jalan Grafika No. 2, Sleman 55281, Indonesia
[c] Biotechnology Master Program, Postgraduate School, Universitas Gadjah Mada, Jalan Teknika Utara, Sleman 55281, Indonesia
[d] Department of Food and Agricultural Product Technology, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia
[e] Bioprocess Engineering Research Group, Center of Excellence Sustainable Environment, Universitas Gadjah Mada, Jalan Grafika No. 2, Sleman 55281, Indonesia

ARTICLE INFO

Editor: Yang Liu

Keywords:
Anaerobic fermentation
Micro aeration
Oil palm empty fruit bunch
Thermophilic
Volatile fatty acids

ABSTRACT

This research studied the conversion of alkaline-pretreated oil palm empty fruit bunch (OPEFB) into volatile fatty acids (VFA) through anaerobic fermentation (AF). In AF, the methanogens that consume VFAs must be suppressed to increase the accumulated VFA as an intermediate product of AF. In-depth research has not been conducted on the effects of temperature and micro aeration to maximize the VFA accumulation. To investigate the temperature effect, the experiment was run at mesophilic (RM) and thermophilic conditions (RT). The impact of the micro aeration dose was studied in the thermophilic reactor set to the ORP −50 mV (RTM −50) and ORP −150 mV (RTM −150). The experimental data were fitted to the Anaerobic Digestion Model 1 (ADM 1) using AQUASIM®. The $k_{hyd.ch}$ value of the RT reactor is higher than that of the RM, suggesting that high temperatures encourage hydrolysis and enhance hydrolase activity. Based on the simulation results, RT reactors without micro-aeration had a substantially higher specific growth rate of methanogens compared to RM. Therefore, micro aeration should be applied in RT to suppress the potentially faster growth of methanogen bacteria at elevated temperature. The result of this study indicated that both temperature and micro-aeration were important process variable to be further optimized to make VFAs production from OPEFB attractive.

1. Introduction

Over the last few decades, palm oil production has grown very rapidly. Consequently, the amount of oil palm empty fruit bunch (OPEFB) as the solid wastes in palm oil industries has increased significantly, posing a severe environmental burden. Based on Indonesia's Central Statistical Bureau (Badan Pusat Statistik/BPS) data, in 2020, Indonesia produced more than 44 million tons of palm oil annually [1]. For every ton of fresh fruit bunches used in the oil extraction process, lignocellulosic biomass was produced, including oil palm empty fruit bunches (OPEFB) (23%), mesocarp fiber (12%), decanter cake (6%), and shell (5%) [2]. Untreated OPEFB contained 15.2% hemicellulose, 25.8% lignin, 40.9% cellulose, and 18.1% miscellaneous components [3]. The rich carbon source in lignocellulosic biomass makes it a highly ideal substrate feedstock for anaerobic processes, which produce carboxylic acids and biogas [4,5]. However, before being used as a substrate, OPEFB must be pretreated because lignin in the biomass forms an impermeable/resistant layer, making it difficult to biodegrade [6]. In another study, pretreated OPEFB could be a source of fermentable sugars [7], such as 2,3-butanediol (2,3-BDO). Other interesting pathway of utilizing lignocellulose waste is anaerobic fermentation (AF) to collect the intermediate product, i.e. volatile fatty acids (VFA) [8,9].

VFAs are low-molecular-weight carboxylates containing 2–6 carbon atoms with primary components of propionic acid, acetic acid, butyric acid, and valeric acid [10]. Some of the potential substrates to produce VFAs include citrus waste [11], activated sludge waste [12], tuna waste [13], household food waste [14], and vegetable waste [15]. VFAs can be utilized as attractive carbon sources for the removal of ammonia, nitrogen, and phosphorus during wastewater treatment [16,17]. They can

* Corresponding author at: Chemical Engineering Department, Universitas Gadjah Mada, Jalan Grafika No. 2, Sleman 55281, Indonesia.
E-mail address: wiratni@ugm.ac.id (W. Budhijanto).

https://doi.org/10.1016/j.jece.2023.110163
Received 29 January 2023; Received in revised form 2 April 2023; Accepted 17 May 2023
Available online 18 May 2023
2213-3437/© 2023 Elsevier Ltd. All rights reserved.

Nomenclature

dS/dt	Rate of substrate consumption (kg_{COD}. m^{-3}.day^{-1}).
f	The yield of product on the substrate (kg_{COD}. kg_{COD}^{-1}).
I_{O2}	Oxygen inhibition function (-).
K	Inhibitory constant ($kg_{COD}.m^{-3}$).
k_{hyd}	First-order constant for hydrolysis (day^{-1}).
k_m	Monod's maximum specific uptake rate (kg_{sCOD}. kg_{xCOD}^{-1}. day^{-1}).
k_s	Half saturation value (kg_{sCOD}. m^{-3}).
n	Number of data points.
p (p_1;.; p_m)	Model parameters for biochemical reactions 1; …; m.
r_{hyd}	Rate of hydrolysis ($kg_{COD}.m^{-3}.day^{-1}$).
$v_{i,j}$	Rate coefficients for component i on process j (kg_{COD}. m^{-3}).
S	Component concentration ($kg_{COD}.m^{-3}$).
X	Particulate component ($kg_{COD}.m^{-3}$).
X^2	Deviations.
Y	The yields of microorganisms mass per unit mass of substrate consumed (kg_{xCOD}. kg_{sCOD}^{-1}).
$y_i(p)$	The model's prediction at the time that data point i occurs, which might be considered a function of the parameter set p that needs to be assessed.
$y_{means,i}$	The i-th measurement's determined value for the measurement target.
$\sigma_{means,i}$	Standard error of measurement.
μ_{max}	Monod's maximum specific growth rate (day^{-1}).

Subscript

ac	Acetic acid.
bu	Butyric acid.
ch	Carbohydrate.
H_2	Hydrogen.
O_2	Oxygen.
pro	Propionate acid.
su	Simple sugar.

also be transformed into various biochemicals required in various industries [18]. VFAs are particularly valuable in the chemical industry as precursors for synthesizing esters, aldehydes, ketones, alcohols, and alkanes [14]. VFAs can also be used to grow a lipid-accumulating oleaginous yeast (i.e., *Apiotrichum porosum* DSM27194) which can further be used as a fuel feedstock for oleochemicals production [19].

Anaerobic fermentation has recently become a cost-effective and environmentally acceptable method of producing VFAs [20–22]. From a metabolic perspective, the AF pathways consist of two main routes, i.e. acidogenic fermentation and methanogenesis [23]. Complex organic polymers are broken down into simple soluble monomers in the first phase of the AF process, called hydrolysis, by extracellular enzymes produced by hydrolytic microbes [24]. Acidogenic bacteria convert the products generated during hydrolysis to volatile fatty acids (VFAs) and minor products such as hydrogen and carbon dioxide (CO_2). Methanogenic bacteria consume the products of acidogenesis and acetogenesis and produce carbon dioxide and methane in the final phase [25]. Therefore, when the AF purpose is to produce VFAs, the challenge is maximizing the accumulation of the VFAs by preventing the consumption of the VFAs by the methanogenes. Meanwhile, increasing the acidification rate is also still challenging [26]. In addition, when dealing with the raw material which is difficult to disintegrade, some special treatments might be necessary, such as the addition of strong oxidizer [17].

Several studies reported the effect of temperature on VFA production from anaerobic fermentation. Garcia-Aguirre et al. [27] concluded that VFA production from organic waste was slightly higher in the process with an acidic pH at thermophilic condition (55 °C). Increasing the temperature and pH accelerated hydrolysis due to increased particulate solubilization [28,29]. To maximize VFA accumulation, the methanogens that consume VFAs should be inhibited [30]. Methanogens are sensitive to environmental variation and various factors, including temperature and micro oxygenation, that could inhibit the activity of methanogens. Recent investigations have shown the benefit of using micro aeration in hydrolytic-acidogenic reactors to dissolve organic substrates, including food waste [31], mixed sludge [32], and Napier grass [33]. Micro aeration is the controlled addition of minute amounts of oxygen to an anaerobic biochemical process. It allows anaerobic and aerobic biological activity in a single bioreactor [34]. Sawatdeenarunat et al. [9] reported the maximum VFAs yield of 107.3 2.6 mg/gVS_{added} was obtained at 15 mL O_2 addition per gVS_{added}, three days of incubation, with anaerobically digested cattle manure (ADCM) as an inoculum. Micro oxygenation may be an environmentally beneficial method to increase the yields of VFAs from lignocellulosic feedstocks for producing high-value chemicals and bioenergy [9]. According to Lim et al. [29], the advantage of micro aeration was to change the fermentation production pattern where acetic acid was converted into butyric acid. Compared to the non-aerated settings, the AF system with micro aeration has more varied bacterial communities. The authors also noted that a micro aerated AF system had a higher percentage of populations from the *Firmicutes* phylum than an anaerobic system, which led to a faster rate of substrate hydrolysis.

Precise oxygen dose control is required to avoid oxygen overdosage. Oxidation-reduction potential (ORP) can be a precise parameter for controlling the proper micro aeration dosage since it measures the net value of all complex oxidation-reduction reactions in an aqueous environment [35]. The redox potential is measured by the ORP, which is also sensitive to oxygen in an aqueous solution [36]. Thus, ORP can be used to define the condition of biochemical reactions [35]. According to Nguyen et al. [37], as soon as the ORP-controlled micro aeration (at ORP of −470 mV) was started, the total VFA concentration quickly dropped to 3.0 g HAc/L, the methane output increased, and oxygen served as the limiting electron acceptor for the facultative heterotrophs. According to Yin et al. [26], the ideal ORP level for VFA synthesis was between − 100 and − 200 mV. Low oxygen resulted in a high yield and higher proportions of facultative acidogenic bacteria (*phylum Firmicutes, Proteobacteria*, etc.).

The entire AF process is better understood with the help of mathematical models, which can also be used to simulate the changes at each process stage [38]. This study evaluated the data using Anaerobic Digestion Model No.1 (ADM1) created by an International Water Association (IWA) work group. The model could increase the understanding of the process configuration, such as the primary biochemical reactions and physico-chemical processes in AF fermentation [39]. The ADM1 is a structural model that includes the processes of disintegration, hydrolysis, acidogenesis, acetogenesis, and methanogenesis. A total of 19 processes, 24 components, and 56 stoichiometric and kinetic parameters are developed for the biological processes involved. As an all-purpose process model, ADM1 can enhance lab-scale and full-scale plant design applications and directly apply them to large-scale plants.

A modified ADM1 has been developed by Bai et al. [40] to simulate substrate degradation and the impact of pH on the production of VFAs. Three substrate degradation models were compared for protein and carbohydrate degradations, including first-order kinetics, i.e., Monod and Contois. The Monod model most accurately approximated the soluble protein and carbohydrate breakdown. The simulated ADM1 by AQUASIM® well predicted the VFA production and the inhibition effects during simulation. A previous study tested oxygen effects in AF

simulated with AQUASIM® [41] using 25 state variables and 22 biochemical processes presented in matrix form. Under the oxygen load of 22, 44 and 88 mg/L, the ADM1-Ox model simulations adequately predicted the experimental methane potentials. The coefficients of determination (R^2) values for the observed data (0.908) and simulated data (0.995) indicated above 90% confidence in data linearity.

This study was aimed to investigate VFA production from OPEFB by increasing temperature and inhibiting methane formation. OPEFB is a complex substrate which needs a combination of elevated temperature and an optimum dose of micro aeration in the AF to maximize the VFA accumulation. The effect of thermophilic temperature and ORP-controlled micro aeration has never been tested on the VFA accumulation optimization and on the microbial population dynamics in the AF of OPEFB substrate. This study was conducted on the AF of alkaline-treated OPEFB. The AF was firstly operated at mesophilic and thermophilic conditions to confirm the effectiveness of the elevated temperature. Secondly, the AF was conducted at the thermophilic conditions with varied micro aeration intensities. The micro aeration level was varied based on the measured ORP. The treatment comparison in the current study was based on ADM1's kinetic constant values. The AQUASIM® software was used to determine the ADM1 kinetics parameters. All parameter values were initially taken from ADM1 defaults for parameters to be estimated. The adjustable parameters are related to the microorganism properties, i.e., $k_{hyd.ch}$, $k_{m.bu}$, $k_{m.pro}$, $k_{m.ac}$, $K_{s.bu}$, $K_{s.pro}$, and $K_{s.ac}$. The differences between the data-fitted constant and the default values will reveal the unique features of the investigated process, which may be necessary for scale-up consideration. The high-throughput 16 s rRNA sequencing was used for microbial identification to cross-check the indications inferred by the model constants from the data fitting on ADM1. The result of this study is useful to recommend the appropriate temperature and ORP values which maximize the VFA accumulation.

2. Material and method

2.1. Inoculum and substrate

Digested cow manure (DCM) was used as the inoculum. The inoculum was taken from the effluent of a mesophilic biogas reactor running on cow manure feedstock in Kaliurang, Yogyakarta, Indonesia. The total plate count (TPC) analysis of the anaerobic bacteria in DCM revealed that the viable cell concentration was 16,000 mg cell.L^{-1}. The oil palm empty fruit bunches (OPEFB) were obtained from the palm oil industry in Medan, North Sumatra Province, Indonesia. Sun-dried OPEFB was chopped and ground into less than 0.42 mm average diameter particles. Untreated OPEFB containing 40.09 ± 0.01%-w glucan, 23.94 ± 0.01%-w hemicellulose, 21.77 ± 0.01%-w lignin, and 3.72 ± 0.07%-w ash. Furthermore, the NREL/TP510–42,620 [42] technique was used to determine the total solid (TS) and volatile solids (VS) of the OPEFB. In addition, the OPEFB was delignified using a chemical method (NaOH, 8% w/v) at 100 ℃ for one hour [43]. Table 1 presents the substrate and inoculum specifications used in this study.

Table 1
Properties of the inoculum (DCM) and the substrate (OPEFB).

Properties	Delignified OPEFB	DCM
VS (gr VS)	12.580	Not determined
VSS (mg.L^{-1})	Not determined	15,680 ± 320
COD (mg.L^{-1})	Not determined	25,025 ± 1025
CODs (mg.L^{-1})	Not determined	931 ± 760
Water (%-w)	85.465 ± 0.135	95.760 ± 0.113
Ash (%-w)	0.030 ± 0	0.625 ± 0.021
Fat (%-w)	0.100 ± 0	0.073 ± 0.020
Total protein (%-w)	0.240 ± 0	0.540 ± 0.014
Total carbohydrate (%-w)	14.255 ± 0.135	3.001 ± 0.168
Cellulose (%-w)	12.967 ± 0.072	Not determined
Hemicellulose (%-w)	1.483 ± 0.023	Not determined
Lignin (%-w)	3.444 ± 0.081	Not determined

2.2. Batch reactor assembly

The pretreated OPEFB and the inoculum with a ratio of 1:1 (VS/VS) were mixed in a beaker glass and put into a 1000 mL batch reactor (Wheaton Spinner Flask, DWK Life Sciences, USA). The amount of the substrate and inoculum mixture was 75% of the reactor volume. The initial pH was set to 5.5 using 1 M NaOH or 1 M HCl. Nitrogen was bubbled into the liquid before the reactor hose was connected to the gasometer. Each test condition was run in duplicates, and the measured parameters were averaged for the final evaluation.

The first set of the experiments were run in mesophilic and thermophilic conditions. The second set of the experiments were carried out in thermophilic condition with and without micro aeration. The micro aeration was controlled at the ORP values of −50 mV and −150 mV. The micro aeration scheme is shown in Fig. 1. The airflow rate from the auto aerator (Jebao, DP-4, China) was set using a digital timer and automatically connected to the probed ORP controller (Mettler Toledo, Columbus, Ohio, USA). The micro aeration intensity (LO_2/L reactor/day) was calculated by multiplying the oxygen flow rate by the daily micro aeration time and dividing it by the working volume of the reactor. A gasometer was connected to the reactor to detect biogas formation. Nevertheless, we did not observe any biogas formation during the study. Therefore, no biogas data was presented.

2.3. Analytical method

Four operating conditions were evaluated in this study (Table 2). Two reactors, i.e., Mesophilic Reactor (RM) and Thermophilic Reactor (RT), were used to study the effect of temperature. Meanwhile, to assess the impact of micro aeration dose, three reactors were used at thermophilic condition, namely the thermophilic reactor without aeration (RT), the thermophilic reactor with aeration at the ORP value of −50 mV (RTM −50), and the thermophilic reactor with aeration at the ORP value of −150 mV (RTM −150). All experimental setups were carried out in duplicates for 16 days. The experiment was stopped when the total sCOD and VFA concentrations did not significantly change anymore.

The parameters observed during the experiment were total chemical oxygen demand (CODt) and soluble chemical oxygen demand (CODs), pH, and VFA concentration. The pH and ORP values are monitored using an electrode (Mettler Toledo, Columbus, Ohio, USA) connected to the reactor system. Samples were centrifuged at 11,000xg for 5 min to separate the fermented broth from the residue. Total solids (TS) and volatile solids (VS) of inoculum and reactor effluent were determined based on the standard analytical procedures by APHA-AWWA-WEF [44]. The CODs and CODt were determined according to the closed reflux method 5220 D [44]. The Hach kit (DR 900, Loveland, Colorado, USA) was used to conduct the CODt and CODs colourimetric analysis. The initial microbial amount of the inoculum was determined using anaerobic total plate count [45]. The 16 s rRNA assay was used to determine the ratio of each variable (Xsu, Xbu, Xpro, Xac) in order to determine the composition of the microbial community.

The VFAs were determined using a gas chromatograph (Perkin-Elmer, Shelton, CT, USA) with a capillary column (Elite-WAX ETR, Perkin-Elmer, Shelton, CT, USA) and a flame-ionized detector (PerkinElmer, Shelton, CT, USA). Injection and detection temperatures were 250 and 300 degrees Celsius, respectively. Nitrogen served as the carrier gas at two mL.min^{-1} flow rate and a pressure of 20 psi. Each injection capacity was 1 μL. Each standard volatile free acid mix (Sigma-Aldrich) was dissolved in water to make a stock solution of caproic acid, isovaleric acid, valeric acid, iso-butyric acid, butyric acid, propionic acid, and acetic acid. A 10 g.L^{-1} mixed stock standard solution was prepared by combining the two solutions.

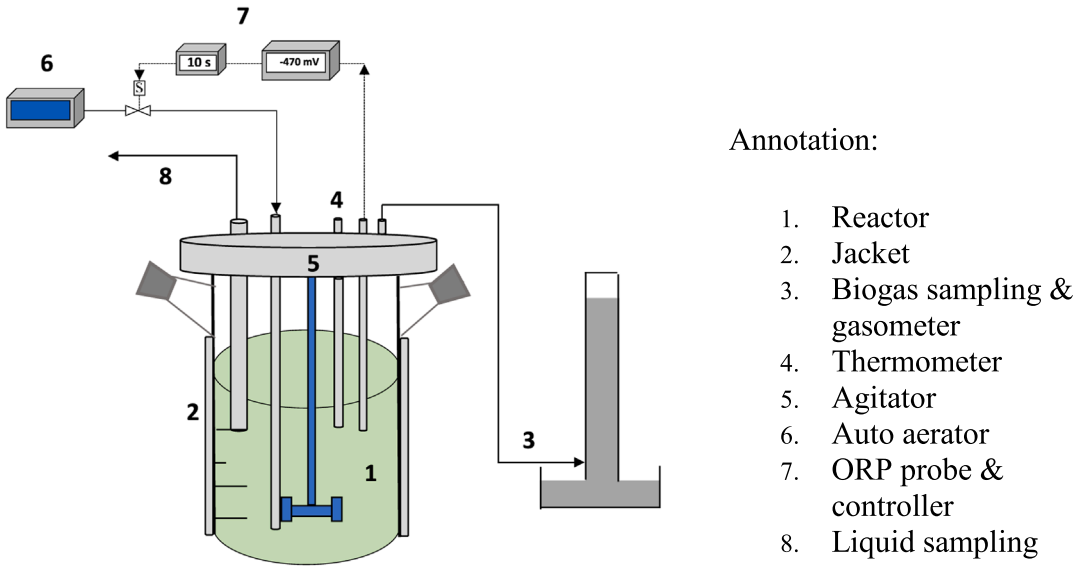


Fig. 1. The batch anaerobic reactor run with temperature and micro aeration variations.

Table 2
Operating conditions for the anaerobic fermentation experiment.

Reactor	ORP (mV)	Oxygen injection	Temperature
RM	±−500	anaerobic	Mesophilic (28 ℃)
RT	±−500	anaerobic	Thermophilic (55 ℃)
RTM-50	-50	487 ± 37 mL.h^{-1}	Thermophilic (55 ℃)
RTM-150	-150	82 ± 27 mL.h^{-1}	Thermophilic (55 ℃)

2.4. Microbial diversity analyses

Digested samples of both liquid and solid phases were collected using a conical tube and immediately frozen at − 20 ℃. The sample was placed at room temperature to melt before running the total DNA analysis. As much as 250 μL of liquid phase sample and 0.5 mg solid-phase was utilized to extract the total microbial DNA with Zymo Research Quick-DNA™ Fecal/Soil Microbe Miniprep Kit D6010 (Zymo Research, United States). DNA isolation was carried out in three stages: lysis, precipitation, and purification/elution. NanoDrop2000 with the dsDNA program was used to determine the concentration and the purity of DNA, while agarose gel electrophoresis was conducted to detect the DNA integrity and fragment size. The 16 S rRNA gene segments were amplified through PCR reactions using the primer 341-F (5′-CCT AYG GGR BGC ASC AG-3′) and 806-R (5′-GGA CTA CNN GGG TAT CTA AT-3′) to both bacteria and archaea and then, sequenced on Illumina NovaSeq 6000 platform (Illumina, United States) by NovogeneAIT Genomics Singapore Pte. (Singapore).

The diversity and species richness indices were calculated using the rRNA Database Project (SILVA database at http://www.arb-silva.de/). Based on the richness estimate of the Simpson [46] and Shannon [47] indices, an OTU-based alpha diversity study was performed to investigate bacterial diversity. The number of sequences divided by the total number of rows per sample was used to determine the relative abundance for dominance hierarchy variables (%).

2.5. Kinetic model for batch reactor

The Anaerobic Digestion Model No. 1 (ADM1) is a model that comprehensively includes all compounds in anaerobic reaction mechanisms validated by International Water Association (IWA) based on various scientific publications and industrial data. Not all aspects of the model always occur in every anaerobic process, so it is necessary to make appropriate assumptions for a particular case. The model's accuracy depends on the accuracy of the assumptions taken in representing specific cases.

The ADM1 distinguishes between biochemical and physicochemical mechanisms in anaerobic fermentation reactions. The first is biologically mediated, describing the four phases of anaerobic fermentation as they relate to the growth and decay of the bacteria engaged in the process. Instead, liquid-liquid, liquid–gas, and acid-base reactions are depicted in the physicochemical model. For each biochemical/chemical compound used in the simulation, the ADM1 allows the user to design a first-order differential equation. This equation depicts the change in concentration of the specific component under consideration over time while accounting for other compounds' concentrations. The differential equation is developed using the Petersen matrices [48], which provide correlations for calculating the kinetic rate equations (expressed with p) and c (specified with v) of each subprocess. for this experiment (batch reactor) it has been considered that every implemented differential equation should be expressed by Eq. (1) (based on the ADM1 equation)

$$\frac{dS_{liq,i}}{dt} = \sum_{j=1-n} p_j v_{i,j} \tag{1}$$

The process scheme modeled with ADM1 in this experiment is presented in Fig. 2. Monod saturation type is used to describe aerobic uptake rates [49]. Experiments were used to identify measurable characteristics such as the composite particle's chemical oxygen demand (COD) and the yield of carbohydrates, lipids, proteins, and inerts from the disintegration of complex particulates. As shown in Eq. (2), the first stage is carbohydrate hydrolysis, in which the carbohydrates in the OPEFB substrate are degraded into simple sugar compounds.

$$r_{hyd,ch} = k_{hyd,ch}.X_{ch}.X_{su} \tag{2}$$

The glucose is transformed into individual volatile fatty acids (VFAs) in the second stage (Acidogenesis), with propionic acid (HPr), butyric acid (Hbu), and acetic acid (HVa) dominating. Eq. (3) depicts the glucose uptake rate at this stage.

$$\frac{dS_{su}}{dt} = \left(-1 + ((1-Y_{SU})f_{bu,SU}\right) + \left((1-Y_{SU})f_{pro,SU}\right) + \left((1-Y_{SU})f_{ac,SU}\right) + \left((1-Y_{SU})f_{h2,SU}\right).\frac{k_{m,su}S_{su}}{k_{s,su}+S_{su}}.X_{su} \tag{3}$$

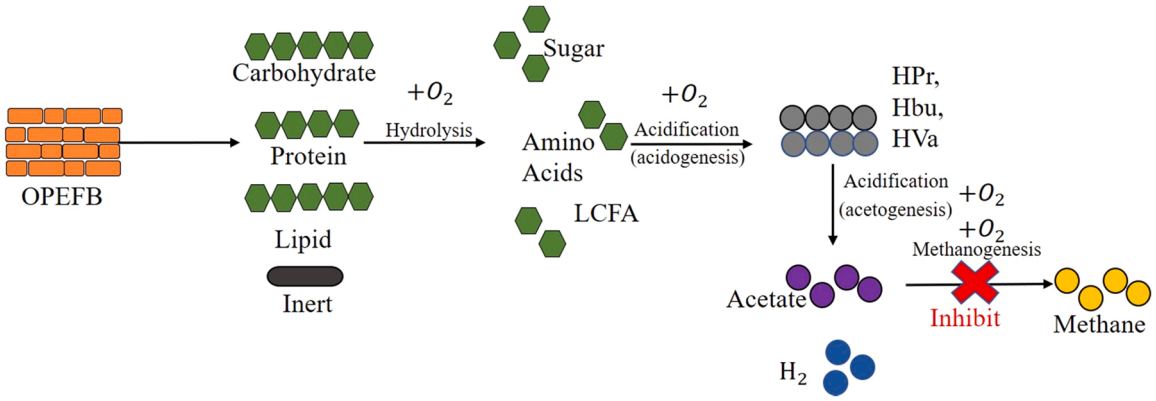


Fig. 2. Schematic process in anaerobic fermentation.

In the third stage, butyric and propionic acid uptake is converted into acetic acid. This study's butyric acid and propionic acid uptake rate can be seen in Eqs. (4) and (5).

$$\frac{dS_{bu}}{dt} = (1 - Y_{C4})0.8 + (1 - Y_{C4})0.2.\frac{k_{m,bu}S_{bu}}{k_{s,bu} + S_{bu}}.X_{bu} \tag{4}$$

$$\frac{dS_{pro}}{dt} = \left(1 - Y_{pro}\right)0.57 + (1 - Y_{SU})0.43.\frac{k_{m,pro}S_{pro}}{k_{s,pro} + S_{pro}}.X_{pro} \tag{5}$$

In the final stage, the acetic acid is transformed into methane, and the acetic acid consumption rate is shown in Eq. (6).

$$\frac{dS_{ac}}{dt} = -1 + (1 - Y_{ac}).\frac{k_{m,ac}S_{ac}}{k_{s,ac} + S_{ac}}.X_{ac} \tag{6}$$

To express the influence of micro aeration by controlling the ORP against the kinetic value constant in anaerobic fermentation, the simulations generated with ADM 1-Ox employ a standard set of kinetic parameters from the ADM 1 model [41]. Dissolved oxygen concentration (O_2) is introduced as a new soluble component, and Eqs. (7–10) show the aerobic uptake rates equation.

$$\frac{dS_{aer,su}}{dt} = \left(-1 + ((1 - Y_{SU})f_{bu,SU}\right) + \left((1 - Y_{SU})f_{pro,SU}\right) + \left((1 - Y_{SU})f_{ac,SU}\right) + \left((1 - Y_{SU})f_{h2,SU}\right).\frac{k_{m,su}S_{su}}{k_{s,su} + S_{su}}.X_{su}.I_{O2} \tag{7}$$

$$\frac{dS_{bu,aer}}{dt} = (1 - Y_{C4})0.8 + (1 - Y_{C4})0.2.\frac{k_{m,bu}S_{bu}}{k_{s,bu} + S_{bu}}.X_{bu}.I_{O2} \tag{8}$$

$$\frac{dS_{pro,aer}}{dt} = \left(1 - Y_{pro}\right)0.57 + (1 - Y_{SU})0.43.\frac{k_{m,pro}S_{pro}}{k_{s,pro} + S_{pro}}.X_{pro}.I_{O2} \tag{9}$$

$$\frac{dS_{ac,aer}}{dt} = -1 + (1 - Y_{ac}).\frac{k_{m,ac}S_{ac}}{k_{s,ac} + S_{ac}}.X_{ac}.I_{O2} \tag{10}$$

The oxygen inhibition effects on the strictly anaerobic acetogenes and methanogens are accounted for in ADM 1-Ox by adding the inhibition term I_{O2}. A well-established non-competitive type inhibitory function (Eq. (11)) was applied as a progressive oxygen switch.

$$I_{O2} = \frac{K_{O2}}{K_{O2} + S} \tag{11}$$

By reducing the sum of the squares of the weighted deviations between measurements and the predicted model outputs, the sensitive parameters identified by the sensitivity analysis were evaluated. The departures are represented by $\chi 2$ formulated in Eq. (12).

$$X^2(p) = \sum_{i=1}^{n}\left(\frac{y_{means,i} - y_i(p)}{\sigma_{means,i}}\right)^2 \tag{12}$$

3. Result and discussion

3.1. VFA production

The profiles of the CODs concentration and the total VFA yield, which reflect OPEFB hydrolysis and acidification at mesophilic and thermophilic conditions, are shown in Fig. 3(a) (RM) and 3(b) (RT). The CODs concentrations in each reactor (RM and RT) showed an upward trend at the beginning of the experiment and then stabilized with the extended fermentation time. The organic content removal is expressed as the total solid (TS) reduction displayed in Table 3. Total solid removal of thermophilic reactors is higher than mesophilic reactors. This data shows that thermophilic condition promotes hydrolysis and improves hydrolase activity [50] to make the acidogenic bacteria more efficiently convert the organic compounds into VFAs. In this study, the total VFA (TVFA) yield was enhanced at a temperature of 55 ℃, and a similar condition was reported by Huang et al. [51], who observed the highest VFA yield at 55 ℃. On the contrary, in the reactors with microaeration, lower TS removal was observed. However, in RTM-150, although the TS removal was lower than RM and RT, the VFA production was similar to RT (Fig. 4). This indicated that the soluble products from TS degradation of OPEFB might be more preferable for acidogenic transformation. The micro-aeration might caused different product of TS degradation because the microbial population especially with the microaeration at −150 mV promoted higher percentage of hydrolytic microorganisms without any methanogenes detected.

In this study, the VFA production was expected to be enhanced by applying micro aeration. The formation of VFAs in each reactor (RM and RT) occurred at the ORP values between −380 mV to −150 mV. Air was injected to maintain the ORP values at −50 mV (RTM-50) and −150 mV (RTM-150). This chosen ORP values were based on previous finding. According to Vongvichiankul et al. [36], the ideal ORP during the acidogenesis phase was −300 mV to −130 mV, and the pH values were 5–7. Yin et al. [52] studied the ORP effect on acidogenic fermentation for VFA production from food waste at four experimental conditions with two ORP levels (−100 to −200 mV).

In the experiment with micro aeration (Fig. 3(c and d)), the highest CODs concentration in RTM-150 occurred on the fifth day and decreased on the seventh day but rose again on the ninth day. The highest CODs concentration in RTM −50 occurred only on the second day and significantly fell until the end of the experiment. From the fourth to the seventh day, the accumulation of VFA in RTM-50 and RTM-150 reactors constantly increased. Each reactor's highest accumulation of VFAs

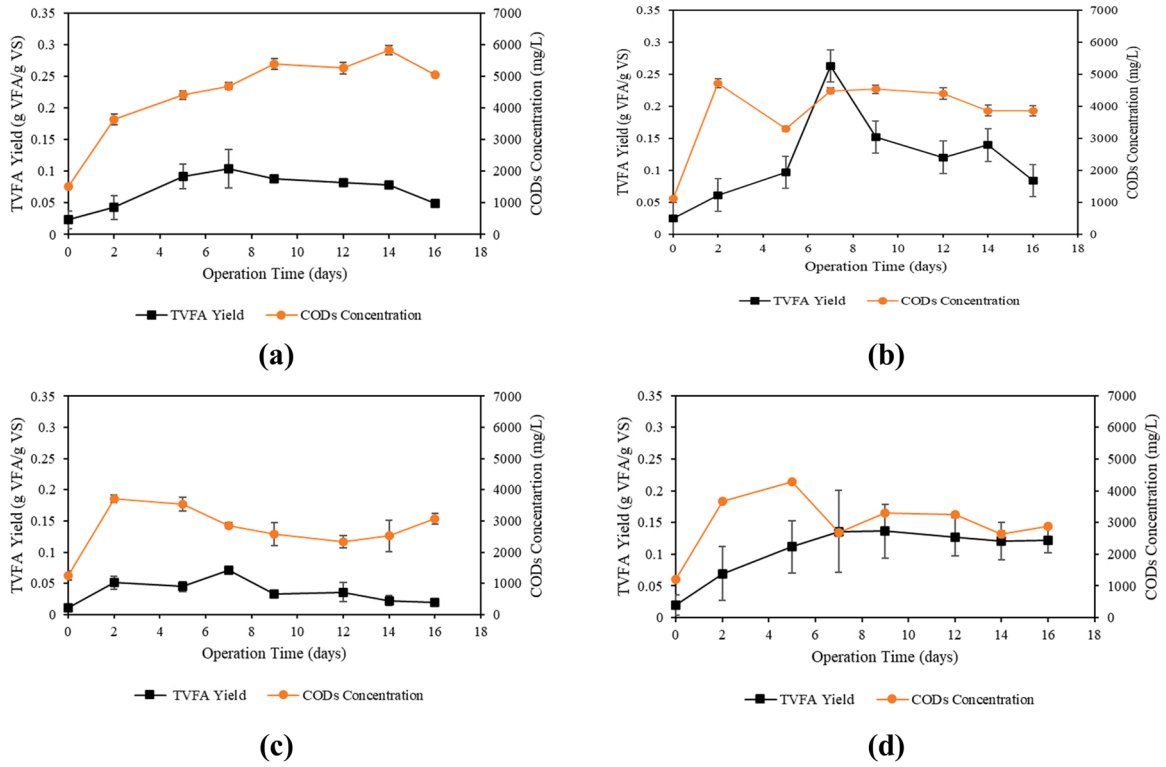


Fig. 3. CODs concentration and total VFA yield; a-b without micro aeration (a) mesophilic (b) thermophilic and c-d with micro aeration (c) ORP −50 mV (d) ORP −150 mV.

Table 3
Resume of overall reactor performance.

Parameter	Reactor RM	Reactor RT	Reactor RTM-50	Reactor RTM-150
Initial pH	5.5	5.5	5.5	5.5
TS at the end of the experiment (mg/L)	23,854 ± 1.388	20,570 ± 4.143	27,560 ± 1.640	27,517 ± 0.759
TS removal (%)	27.933	37.855	16.737	16.865

occurred on the seventh day (Fig. 3(c and d)). The highest yield of VFAs in the RTM −50 reactor and the RTM −150 were 0.072 ± 0. $g_{VFA}.g_{VS}^{-1}$ and 0.136 ± 0.065 $g_{VFA}.g_{VS}^{-1}$, respectively, the lowest accumulation of VFAs occurs in reactors when there was excessive aeration, namely at ORP −50 mV. Considering the CODs concentration profile in Fig. 3(c), the RTM −50 CODs concentration was lower than those in the other three reactors. This observation indicated that at some point, micro aeration inhibited the metabolism of hydrolytic microorganisms, reducing the rate of organic compound dissolution, and producing less VFAs. The airflow rate in RTM −50 was 47% higher than in the RTM −150 reactor. Undeniably, anaerobic reactors without micro aeration treatment (RT) produced the highest accumulation of VFAs compared to RTM −50 and RTM −150 (Fig. 3(c and d)). Based on the results above, micro aeration did not seem advantageous to the hydrolysis and acidification processes.

The individual VFAs in each reactor, i.e. acetate, propionate, and butyrate, are shown in Fig. 4. The most dominant individual VFA in all reactors was acetic acid, with the highest accumulation observed on the seventh day. The composition of the VFA constituents in this study was similar to the published results in Table 4.

Butyrate and propionate were found in all reactors. The butyrate accumulation in the RT reactor was higher than that of RM. In contrast, the accumulation of propionate was higher in RM. The highest accumulation of butyrate and propionate occurred on the seventh day (Fig. 4 (a and b)), which was 124.235 ± 128 mg. L^{-1} for RM and 345.832 ± 123 mg. L^{-1} for RT. The composition of propionate is 468.179 ± 142 mg. L^{-1} for RM and 77.042 ± 45 mg. L^{-1} to RT. The increased accumulation of VFAs coincided with the reactor's increased concentration of butyrate and propionate. In general, butyric and propionic acids have a minimal role in anaerobic fermentation. Acetic acid formation reduces the production of butyrate and propionate [53]. However, a low pH value can trigger the presence of butyric and propionic acids.

Acetate, propionate, n-butyrate, iso-butyrate, valerate, and iso-valerate were some of the individual variances in VFAs discovered in reactors with micro aeration, as presented in Fig. 4(c and d). In all reactors, the dominant VFA was acetic acid, i.e., 92.5% in RTM-50% and 81.2% in RTM −150. In RTM −50, the second highest individual accumulation after acetate is propionate. The peak of propionate and butyrate concentration in RTM −50 was observed on the fifth day (41.669 ± 21 mg. L^{-1} and 47.204 ± 16 mg. L^{-1}). The highest propionate concentration in RTM −150 was obtained on the seventh day (162.436 ± 101 mg. L^{-1}), and the highest butyrate concentration on the ninth day (119.819 ± 81 mg. L^{-1}). The presence of n-butyric, iso-butyric, valeric, and iso-valeric acids only happened in RTM −50 and RTM −150, indicating that micro aeration disturbed the anaerobic process. This abnormal condition is favorable when the more diversed VFA composition is the desired product.

3.2. Microbial community

Samples for microbial community analysis from all reactors (RT, RM, RTM-50, and RTM-150) were collected on the seventh day when the VFA production was the highest. The 16 S rRNA high-throughput sequencing was performed to evaluate the microbial community contribution to the reactor performance. Table 5 shows the sequence and taxon numbers of bacteria and archaea. The abundance-based Chao1 indices showed that the microbial abundance in the inoculum (DCM) was higher than mesophilic, thermophilic, and micro aeration

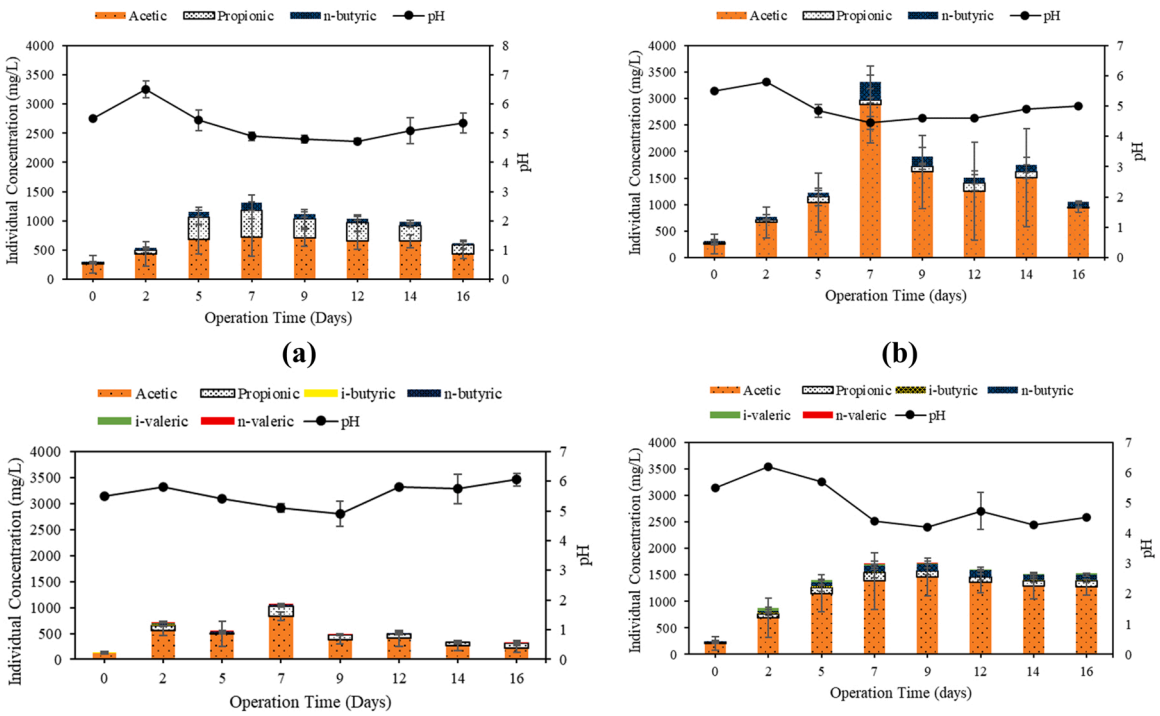


Fig. 4. Individual VFA concentration; a-b without micro aeration (a) mesophilic (b) thermophilic and c-d with micro aeration (c) ORP −50 mV (d) ORP −150 mV.

Table 4
VFA production from several forms of fermentation.

Substrates	Digestion			Performance of VFA production	VFA Composition	Ref.
	Type	**Time**	**Aeration**			
Tuna waste	Mesophilic (37 °C)	40 days	No aeration	30,611 mg COD/L	Acetic, propionic, n-butyric, and i-valeric acids	[13]
Organic Waste	Thermophilic (55 °C)	10 days	No aeration	677 mg COD g−1 CODfed	Acetic, butyric, valeric, propionic, i-valeric, and i-valeric acids	[27]
Synthetic food waste (FW)	Mesophilic (35 °C)	18 days	Micro aeration: 1.0 L/min	0.39 g COD/g VS added	Acetic, butyric, n- butyric, and propionic acids	[54]
Simulated food waste	Mesophilic (30 ± 2 °C)	21 days	Micro aeration: ORP level −100 to −200 mV	0.79 g COD/g VS	Acetic, propionic, iso-butyric, n-butyric, i-valeric, and n- valeric acids	[52]
Oil Palm Empty Fruit Bunch (OPEFB)	• Mesophilic 28 °C (RM) • Thermophilic 55 °C (RT) • Thermophilic with micro-aeration ORP −50 mV (RTM −50) • Thermophilic with micro-aeration ORP −150 mV (RTM −150)		Without and with micro aeration (ORP −50 and −150 mV)	• 0.104 ± 0.031 $g_{VFA}.g_{VS}^{-1}$ (RM) • 0.263 ± 0.081 $g_{VFA}.g_{VS}^{-1}$ (RT) • 0.072 ± 0.002 $g_{VFA}.g_{VS}^{-1}$ (RTM −50) • 0.136 ± 0.065 $g_{VFA}.g_{VS}^{-1}$ (RTM −150)	Acetate, propionate, and butyrate	This study

Table 5
Alpha diversity indexes of microbial communities.

Sample ID	Shanon	Simpson	Chao1 index
Inoculum (Initial) without OPEFB	7.310	0.968	2128.927
RM	7.134	0.979	1504.595
RT	7.191	0.980	1396.789
RTM-50	3.034	0.740	1037.965
RTM-150	3.140	0.722	1019.438

treatment (RTM-50 and RTM-150) in AD systems. The community abundances of inoculum decreased after fermentation because acidic and anaerobic fermentation conditions temporarily inhibit hetero-bacteria [55,56]. Shannon and Simpson's indexes were employed to assess the bacteria diversities. As shown in Table 5, both Shannon and Simpson indices were decreased, indicating that the bacterial diversity of RT, RM, RTM-50, and RTM-150 was lower than in DCM [57]. The Shannon diversity index of RTM-50 and RTM-150 samples slightly decreased compared to inoculum, RM, and RTM with an increment in VFA production. It means that microbial diversity decreased in RTM-50 and RTM-150 because the system was gradually dominated by the acid-producing bacteria, i.e. *Thermoanaerobacterium* and *Romboutsia*. The microbial dominance shift happened because micro aeration led to the selection of facultative microorganisms, and adaptable functional flora became gradually enriched during the process. The decrease in diversity is also due to a decrease in the percentage of archaea. As shown in Fig. 5(a), the archaea had a relative abundance of 16.69% (*Methanothermobacter*) in the inoculum sample, 2.34% (*Methanosaeta*) in the RM reactor, and 8.91% (*Methanosaeta*) in the RT reactor. Archaea were not detected throughout the experimental period in samples RTM-50

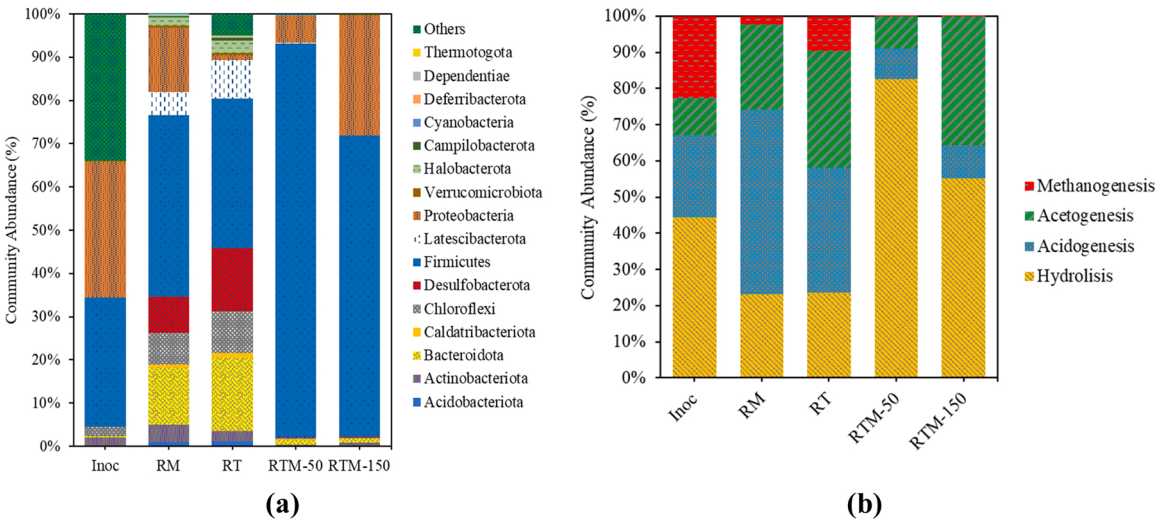


Fig. 5. Relative abundance changes throughout the process, (a) at the phylum level and (b) microbial community.

and RTM-150, thereby indicating successful inhibition of the methanation process during long-term VFA production at ORP −50 and −150 mV.

The major microbial groupings at the phylum level (Fig. 5(a)) were *Bacteroidota, Chloroflexi, Desulfobacterota, Firmicutes, and Proteobacteria*. Anaerobic fermentation to produce VFA promoted the richness of *Bacteroidota,* which increased from 0.39% in inoculum to 13.14% (RM), 16.90% (RT), 1.21% (RTM-50), and 0.85% (RTM-150). The abundance of *Firmicutes* in the inoculum was 45.22% and significantly increased to 91.25% at RTM-50% and 69.66% at RTM-150. *Firmicutes* and *Bacteroidota* are two common acid-forming bacteria that break down cellulose and make organic acids as metabolic byproducts [8,58,59]. *Bacteroidota* can degrade organic materials and produce lytic enzymes and acetic acid [58].

Bacteriodota is commonly found in completely anaerobic conditions, contrary to the *Firmicutes* which is commonly found in reactors with micro aeration. A phylum of facultative bacteria called *Firmicutes* produces extracellular enzymes such as cellulases, lipases, and proteases [52]. *Firmicutes* were highest detected at an ORP −50 mV and −150 mV, because of the potent inhibition of methanogenic activity under micro aeration conditions. On the contrary, without aeration condition, Phylum *Chloroflexi* and *Desulfobacterota* were the highest, with the composition of *Chloroflexi* comprising 7.39% in RM and 9.72% in RT, while *Desulfobacterota made* 8.34% in RM and 14.63% in RT. It might be speculated that *Chloroflexi* and *Desulfobacterota* owned a weak tolerance for oxygen presence in the reactor. The abundance of other bacteria increased due to high temperature, decreasing *Chloroflexi* and *Desulfobacterota* abundance. Meanwhile, the abundance of *Proteobacteria* decreased significantly, from 47.63% in inoculum to 14.92% (RM), 1.29% (RT), 6.34% (RTM-50), and 27.81% (RTM-150). Besides, Yin et al. [52] also found a similar phenomenon in limited aeration to produce VFA.

Fig. 5(b) shows four prominent roles in the microorganism classification, i.e., hydrolytic, acidogenic, acetogenic, and methanogenic. The microbial community in the inoculum consisted of hydrolytic microbes (44.44%), acidogenic microbes (22.75%), acetogenic microbes (10.23%), and methanogenic microbes (22.59%). In the mesophilic condition (RM), the microbial community was comprised of hydrolytic microbes 23.19% (*Christensenellaceae_R-7_group*), acidogenic microbes 51.04% (*Acinetobacter*), acetogenic microbes 33% (*Proteiniclasticum*), and methanogenic microbes 2.39% (*Methanosaeta*). The dominant microbes were acidogenic microbial. There was a significant change in the percentage of acidogenic microbes in the reactor samples compared to the inoculum. In RM reactors, the acidogenic microbes percentage was twice higher than in the inoculum. This phenomenon is caused by a rise in the number of acidogenic microbes and a rise in the quantity of VFA production in mesophilic, inhibiting the growth of other microbes. The microbial composition in the thermophilic reactor was similar to the mesophilic one. On the RT reactor, the microbial composition is 23.72% of hydrolytic microbes (*Christensenellaceae_R-7_group*), 34.54% of acidogenic microbes (*Latescibacterota*), 32.04% of acetogenic microbes (*DMER64*), and 9.71% of methanogenic microbes (*Methanosaeta*).

On the contrary, micro aeration changed the microbial community. In RTM-50, the microbial community consisted of 82.58% hydrolytic microbes (*Incertae_Sedis*), 8.49% acidogenic microbes (*Thermoanaerobacterium*), 8.90% acetogenic microbes (*Tepidiphilus*), and 0.03% methanogenic microbes (*Methanosaeta*). The microbial community on RTM-150 consisted of 55.25% hydrolytic microbes (*Incertae_Sedis*), 9.07% acidogenic microbes (*Romboutsia*), and 35.08% acetogenic microbes (*Tepidiphilus*), and no methanogenic microbes. Microaeration promotes the growth of hydrolytic microbes. Under aerobic conditions, hydrolysis rates are shown to be significantly higher, perhaps due to increased enzyme synthesis. According to previous studies [26,31], a micro aeration environment could stimulate the growth of hydrolytic (facultative) bacteria.

No methanogenic microbes are found in reactors with micro aeration. This finding shows that keeping the ORP value in the range of −50 to −150 mV can inhibit the growth of methanogenic microbes. Based on the 16 s rRNA assay in this study, fermentative (acidogenic and acetogenic organisms) are commonly found in reactors with ORP at −150 mV. This tendency shows that fermentative microbes can still adapt to slight micro aeration and will be inhibited in growth at extreme micro aeration.

3.3. Simulation and kinetic parameter

Kinetic analysis of the anaerobic fermentation process at temperature variations and micro aeration effects was carried out by fittings the experimental data on the ADM1 model described in Subsection 2.5.2 using the AQUASIM® program. The fitting result of individual VFA (acetate, propionate, and n-butyrate) data is shown in Fig. 6. The ADM1 and AQUASIM® are most ideally applied to liquid systems with excellent solubility of compounds. This study used a solid substrate, so the model-fitting results were not as perfect as the usual model fittings under ideal solution conditions. However, the simulated results were close to the experimental data. Hence, the values of the adjustable kinetic constants could be used as the comparison tool to evaluate the treatments in this study. The values of the kinetic constants are summarised in Table 6.

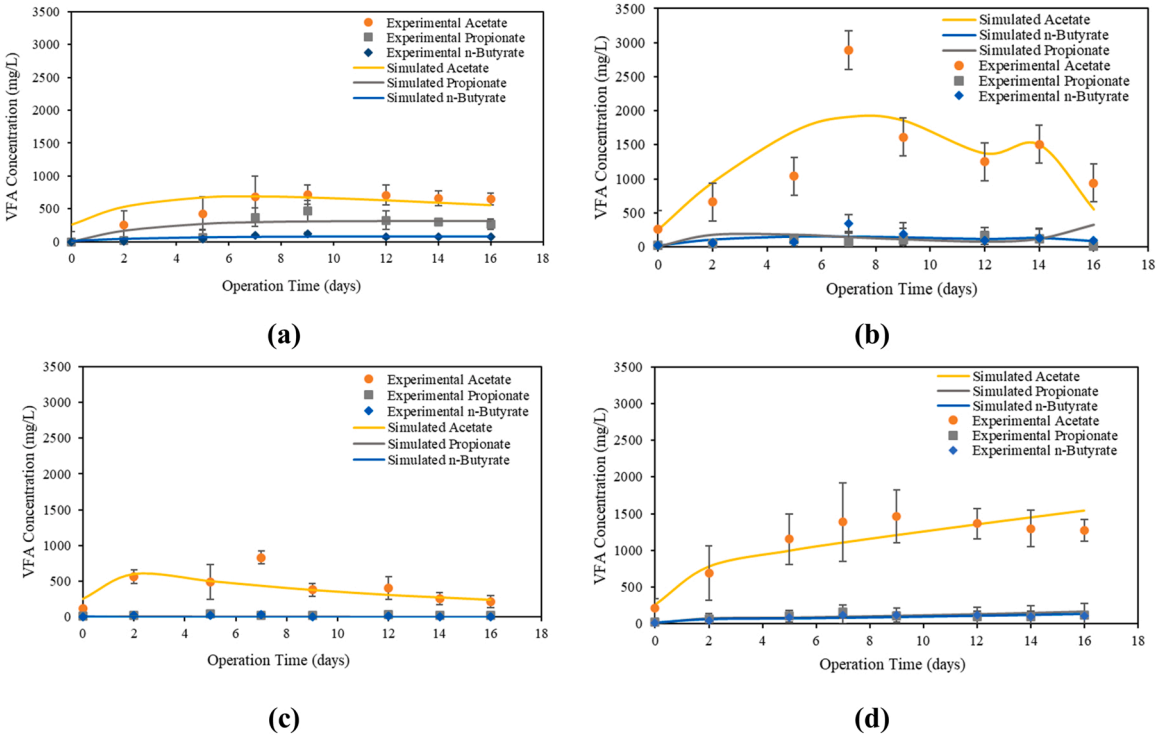


Fig. 6. Results of VFA simulation using AQUASIM®.

Table 6
Comparison of the fitted parameter from this study with a proposed parameter value.

Parameter	Fitted Parameter				Suggested parameter values on mesophilic solids [48]	Suggested parameter values on Thermophilic solids [48]
	RT	RM	RTM-50	RTM-150		
$k_{hyd.ch}$ (day^{-1})	0.224	0.038	0.042	0.043	10	10
$k_{m.bu}$ ($kg_{sCOD}.kg^{-1}_{xCOD}.day^{-1}$)	60.595	5.189	17.129	1.892	20	30
$k_{m.pro}$ ($kg_{sCOD}.kg^{-1}_{xCOD}.day^{-1}$)	15.285	1.406	7.227	0.926	13	20
$k_{m.ac}$ ($kg_{sCOD}.kg^{-1}_{xCOD}.day^{-1}$)	0.033	0.026	1.428	1.006	8	16
$K_{s.su}$ ($kg_{sCOD}.m^{-3}$)	90.691	0.500	0.010	1.000	0.5	1.000
$K_{s.bu}$ ($kg_{sCOD}.m^{-3}$)	38.489	5.189	2.995	7.978	0.2	0.4
$K_{s.pro}$ ($kg_{sCOD}.m^{-3}$)	15.475	13.977	2.967	1.684	0.1	0.3
$K_{s.ac}$ ($kg_{sCOD}.m^{-3}$)	1.119	3.995	2.902	9.992	0.15	0.3
Y_{ac} ($kg_{xCOD}.kg^{-1}_{sCOD}$)	9.517	0.254	0.071	0.050	0.05	0.05

According to the previous studies [40,60,61], the most possibly significant parameters affecting the output of the kinetic model were the hydrolysis rate for carbohydrates ($k_{hyd,\ ch}$), half-saturation value for butyrate ($k_{s,\ bu}$), propionate ($k_{s,\ pro}$), and Monod maximum specific uptake rate for butyrate and propionate ($k_{m,\ bu}$ and $k_{m,\ pro}$) in the acidogenesis/acetogenesis phase. The kinetic parameters of k_{hyd} indicate the substrate's decomposition rate in the anaerobic reactor [48]. The greater the value of $k_{hyd,}$ the faster the breakdown and solubilization of the complex organic material to soluble substrates. Table 6 shows that the RT reactor had a higher $k_{hyd.ch}$ value than the RM, RTM-50, and RTM-150 reactors. The magnitude of the k_{hyd} value of the RT reactor indicated that thermophilic conditions successfully increased the solubility of organic compounds from OPEFB. RTM-50 and RTM-150 reactors were also operated under thermophilic conditions, and the hydrolysis did not significantly increase the k_{hyd} values compared to RT reactors although the hydrolytic microbial community shown in Fig. 5 (b) was much higher in RTM −50 and RTM −150. This phenomenon indicates that the presence of oxygen to maintain ORP in the range of −50 to −150 mV can increase the growth rate of hydrolytic microorganisms but did not significantly affect the hydrolytic rate.

The maximum specific growth rate (μ_{max}) is one of the essential parameters in the kinetics of the AF process that shows how fast the maximum growth of microorganisms at each stage is. The μ_{max} value of methanogen microorganisms is obtained by multiplication of $k_{m,ac}$ and the yield of soluble acetate (Y_{ac}), which resulted in the μ_{max} value of RM reactor of 0.254 day^{-1}, RT reactor 0.315 day^{-1}, RTM-50 reactor of 0.071 day^{-1}, and RTM-150 reactor of 0.0503 day^{-1}. The μ_{max} value of RT reactors is much greater than that of RM reactors. Temperature increase significantly affected the growth rate and metabolism of methanogen microorganisms. Therefore, Fig. 5(b) showed that the communities of methanogenic microorganisms in RT reactors were higher when compared to RM reactors. With micro aeration, the μ_{max} value of methanogen microorganisms was much lower than anaerobic reactors without micro aeration. This phenomenon indicated that the presence of oxygen with ORP values of −50 to −150 mV inhibited methanogenic growth. This statement is also supported by the microbial community analysis shown in Fig. 5(b). No methanogen microorganisms were found in the RTM-50 and RTM-150 reactors.

Another critical parameter is the K_S value, which expresses the substrate concentration necessary to achieve half of the maximum growth rate of the acidogenic microorganism (μ_{max1}). The values of $K_{s,su}$, $K_{s,bu}$, and $K_{s,pro}$ also indicate the bacterial-substrate affinity in the OPEFB degradation during the acidogenesis/acetogenesis stage. The low K_S value indicates good bacterial-substrate affinity. Table 6 shows that the results of the fitted parameters $K_{s,su}$, $K_{s,bu}$, dan $K_{s,pro}$ in RTM-50 and RTM −150 reactors were significantly different than RT reactors. Table 6 shows that in the reactors with micro aeration, the hydrolysis stage of carbohydrates becomes sucrose (su), and the consumption of propionate and butyrate into acetate is the rate-limiting (slowest) so that the acidogenic process became the determinant of the overall rate of AF. Meanwhile, in RT reactors, the value of $K_{s,ac}$ was lower than three other kinetic parameters, so that at the stage of methanogenesis, it becomes rate-limiting in the AF process of RT reactors. Optimization of anaerobic fermentation time is needed when using thermophilic conditions without micro aeration before acetate is converted to methane gas.

Based on the constants associated with the acidogenic process, reactors with micro aeration increased VFA production more than reactors without micro aeration. Micro aeration does not inhibit the growth rate of hydrolytic and acidogenic microorganisms but affects the metabolism in producing VFA. This statement is also supported by the community analysis of microorganisms shown in Fig. 5(b). Overall, other optimization methods are still needed to increase the metabolism of acidogenic microorganisms. Therefore, micro aeration optimization is needed to maximize the process of forming VFAs at the stage of acidogenesis.

4. Conclusion

In summary, this work study the effect of elevated temperature and micro-aeration on the amount of accumulated VFAs in the AF of alkaline-pretreated OPEFBs. It was found that: 1) the thermophilic condition benefited hydrolysis and enhanced hydrolase activity as shown by the higher $k_{hyd.ch}$ value in RT than in RM. On the other hand, the hydrolysis and acidification processes did not benefit from micro aeration, 2) The hydrolytic microbial population was substantially more dominant in the RTM-50 and RTM-150 reactors compared to RT, the hydrolysis rates did not look appreciably different from those in RT reactors, 3) Mesophilic condition favored more buildup of propionate. In contrast, higher temperatures led to the accumulation of longer fatty acids, besides the higher production of acetic acid. The buildup of butyrate was more significant in the RT reactor than in the RM reactor. More diverse VFA individuals were discovered in reactors with micro aeration in the RT. N-butyric, iso-butyric, valeric, and iso-valeric acids were only found in RTM −50 and RTM −150, 4) The selection of facultative microorganisms was favored by micro aeration, and an adaptive functional flora was gradually enriched as a result. The absence of methanogenic bacteria in the reactors with micro aeration demonstrated that keeping the ORP value between −50 and −150 mV inhibited the methanogens, 5) Based on the simulation results, RT reactors without micro-aeration induced higher specific growth rate of methanogens compared to RM. Therefore, while the experimenal data showed that micro-aeration did not positively affect the hydrolysis and acidogenesis rate, micro aeration at an optimized dosage should be applied in RT to suppress the potentially rapid growth of methanogen bacteria at elevated temperature.

CRediT authorship contribution statement

Andri Sanjaya: Investigation, Visualization, Formal analysis, Writing – original draft; **Deni Frans Sakka:** Investigation; **Rara Ayu Lestary:** Data curation, Formal analysis; **Roberty Alvania Kartika:** Investigation; **Desi Riana Saputri:** Writing – review; **Damayanti Damayanti:** writing – review; **Ria Millati:** Conceptualization, Funding acquisition, Project administration, Writing – review; **Wiratni Budhijanto:** Supervision, writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors appreciate the research grant from the Ministry of Research and Technology/National Research and Innovation Agency of the Republic of Indonesia (RISTEK-BRIN) (grant number 3194/UN1. DITLIT/DIT-LIT/PT/2020) and Institut Teknologi Sumatera (grant number B/763 y/IT9. C1/PT.01.03/2022). We also thank Mr. Yudi Iskandarsyah for proofreading the manuscript, Mr. Daniel Tanto for the reactor installation and maintenance, and Mr. Achmad Saifudin for the technical assistance of the laboratory work.

References

[1] B.P.S., Statistik Kelapa Sawit Indonesia, Available Online. (n.d.). Badan Pus. Stat. (Accessed August 14, 2022). (2022). Badan Pusat Statistik (bps.go.id) (accessed August 14, 2022).
[2] W. Suksong, P. Kongjan, P. Prasertsan, T. Imai, S. O-Thong, Optimization and microbial community analysis for production of biogas from solid waste residues of palm oil mill industry by solid-state anaerobic digestion, Bioresour. Technol. 214 (2016) 166–174, https://doi.org/10.1016/j.biortech.2016.04.077.
[3] L. Darus, S. Susana, H. Sihombing, A.R.I. Utami, M. Mel, Enzymatic hydrolysis enhancement of oil palm empty fruit bunch by Peracetic-Sulfuric acid pretreatment, Chem. Eng. J. 429 (2022), 132452, https://doi.org/10.1016/j.cej.2021.132452.
[4] H. Zou, Q. Jiang, R. Zhu, Y. Chen, T. Sun, M. Li, J. Zhai, D. Shi, H. Ai, L. Gu, Q. He, Enhanced hydrolysis of lignocellulose in corn cob by using food waste pretreatment to improve anaerobic digestion performance, J. Environ. Manag. 254 (2020), https://doi.org/10.1016/j.jenvman.2019.109830.
[5] S. Paul, A. Dutta, Challenges and opportunities of lignocellulosic biomass for anaerobic digestion, Resour. Conserv. Recycl. 130 (2018) 164–174, https://doi.org/10.1016/j.resconrec.2017.12.005.
[6] X. Liu, D. Wang, Z. Chen, W. Wei, G. Mannina, B.J. Ni, Advances in pretreatment strategies to enhance the biodegradability of waste activated sludge for the conversion of refractory substances, Bioresour. Technol. 362 (2022), 127804, https://doi.org/10.1016/j.biortech.2022.127804.
[7] Z. Zanirun, E.K. Bahrin, P. Lai-Yee, M.A. Hassan, S. Abd-Aziz, Enhancement of fermentable sugars production from oil palm empty fruit bunch by ligninolytic enzymes mediator system, Int. Biodeterior. Biodegrad. 105 (2015) 13–20, https://doi.org/10.1016/j.ibiod.2015.08.010.
[8] X. Zuo, H. Yuan, A.C. Wachemo, X. Wang, L. Zhang, J. Li, H. Wen, J. Wang, X. Li, The relationships among sCOD, VFAs, microbial community, and biogas production during anaerobic digestion of rice straw pretreated with ammonia, Chin. J. Chem. Eng. 28 (2020) 286–292, https://doi.org/10.1016/j.cjche.2019.07.015.
[9] C. Sawatdeenarunat, S. Sung, S.K. Khanal, Enhanced volatile fatty acids production during anaerobic digestion of lignocellulosic biomass via micro-oxygenation, Bioresour. Technol. 237 (2017) 139–145, https://doi.org/10.1016/j.biortech.2017.02.029.
[10] S. Wainaina, M. Lukitawesa, Kumar Awasthi, M.J. Taherzadeh, Bioengineering of anaerobic digestion for volatile fatty acids, hydrogen or methane production: a critical review, Bioengineered 10 (2019) 437–458, https://doi.org/10.1080/21655979.2019.1673937.
[11] B. Eryildiz, M.J. Lukitawesa, Taherzadeh, Effect of pH, substrate loading oxygen and methanogens inhibitors on volatile fatty acid (VFA) production from citrus waste by anaerobic digestion, Bioresour. Technol. 302 (2020), https://doi.org/10.1016/j.biortech.2020.122800.
[12] T. Liang, K. Elmaadawy, B. Liu, J. Hu, H. Hou, J. Yang, Anaerobic fermentation of waste activated sludge for volatile fatty acid production: recent updates of pretreatment methods and the potential effect of humic and nutrients substances, Process Saf. Environ. Prot. 145 (2021) 321–339, https://doi.org/10.1016/j.psep.2020.08.010.
[13] N. Bermúdez-Penabad, C. Kennes, M.C. Veiga, Anaerobic digestion of tuna waste for the production of volatile fatty acids, Waste Manag. 68 (2017) 96–102, https://doi.org/10.1016/j.wasman.2017.06.010.
[14] G. Strazzera, F. Battista, M. Andreolli, M. Menini, D. Bolzonella, S. Lampis, Influence of different household food wastes fractions on volatile fatty acids production by anaerobic fermentation, Bioresour. Technol. 335 (2021), 125289, https://doi.org/10.1016/j.biortech.2021.125289.

[15] Q. Zhang, Y. Lu, X. Zhou, X. Wang, J. Zhu, Effect of different vegetable wastes on the performance of volatile fatty acids production by anaerobic fermentation, Sci. Total Environ. 748 (2020), https://doi.org/10.1016/j.scitotenv.2020.142390.
[16] H. Lin, Y. Chen, N. Shen, Y. Deng, W. Yan, R. Ruhyadi, G. Wang, Effects of individual volatile fatty acids (VFAs) on phosphorus recovery by magnesium ammonium phosphate, Environ. Pollut. 261 (2020), 114212, https://doi.org/10.1016/j.envpol.2020.114212.
[17] Q. Xu, Q. Fu, X. Liu, D. Wang, Y. Wu, Y. Li, J. Yang, Q. Yang, Y. Wang, H. Li, B. J. Ni, Mechanisms of potassium permanganate pretreatment improving anaerobic fermentation performance of waste activated sludge, Chem. Eng. J. 406 (2021), 126797, https://doi.org/10.1016/j.cej.2020.126797.
[18] K. Wang, J. Yin, D. Shen, N. Li, Anaerobic digestion of food waste for volatile fatty acids (VFAs) production with different types of inoculum: Effect of pH, Bioresour. Technol. 161 (2014) 395–401, https://doi.org/10.1016/j.biortech.2014.03.088.
[19] X. Qian, X. Zhou, L. Chen, X. Zhang, F. Xin, W. Dong, W. Zhang, K. Ochsenreither, M. Jiang, Bioconversion of volatile fatty acids into lipids by the oleaginous yeast Apiotrichum porosum DSM27194, Fuel 290 (2021), 119811, https://doi.org/10.1016/j.fuel.2020.119811.
[20] L. Feng, Y. Chen, Enhancement of waste activated sludge protein conversion and volatile fatty acids accumulation during waste activated sludge anaerobic fermentation by carbohydrate substrate addition: the effect of pH, Environ. Sci. Technol. (2009) 4373–4380.
[21] J. Zhao, D. Wang, X. Li, Q. Yang, H. Chen, Y. Zhong, G. Zeng, Free nitrous acid serving as a pretreatment method for alkaline fermentation to enhance short-chain fatty acid production from waste activated sludge, Water Res. 78 (2015) 111–120, https://doi.org/10.1016/j.watres.2015.04.012.
[22] X. Liu, M. Du, J. Yang, Y. Wu, Q. Xu, D. Wang, Q. Yang, G. Yang, X. Li, Sul fi te serving as a pretreatment method for alkaline fermentation to enhance short-chain fatty acid production from waste activated sludge, Chem. Eng. J. 385 (2020), 123991, https://doi.org/10.1016/j.cej.2019.123991.
[23] F. Valentino, G. Munarin, M. Biasiolo, C. Cavinato, D. Bolzonella, P. Pavan, Enhancing volatile fatty acids (VFA) production from food waste in a two-phases pilot-scale anaerobic digestion process, J. Environ. Chem. Eng. 9 (2021), https://doi.org/10.1016/j.jece.2021.106062.
[24] M.-H. Cui, L. Chen, X.-D. Zhang, Q. Zhang, H. Pan, L.-Y. Liu, H. Liu, A.-J. Wang, Recent advancements on the migration and transformation of hydrophobic pharmaceutically active compounds in anaerobic digestion process, Chem. Eng. J. 446 (2022), 136902, https://doi.org/10.1016/j.cej.2022.136902.
[25] S.N. Ugwu, K. Harding, C.C. Enweremadu, Comparative life cycle assessment of enhanced anaerobic digestion of agro-industrial waste for biogas production, J. Clean. Prod. 345 (2022), 131178, https://doi.org/10.1016/j.jclepro.2022.131178.
[26] X. Huang, C. Duan, J. Yu, W. Dong, H. Wang, Response of VFAs and microbial interspecific interaction to primary sludge fermentation temperature, J. Clean. Prod. 322 (2021), 129081, https://doi.org/10.1016/j.jclepro.2021.129081.
[27] J. Garcia-Aguirre, E. Aymerich, J. González-Mtnez. de Goñi, M. Esteban-Gutiérrez, Selective VFA production potential from organic waste streams: assessing temperature and pH influence, Bioresour. Technol. 244 (2017) 1081–1088, https://doi.org/10.1016/j.biortech.2017.07.187.
[28] E. Hosseini Koupaie, L. Lin, A.A. Bazyar Lakeh, A. Azizi, B.R. Dhar, H. Hafez, E. Elbeshbishy, Performance evaluation and microbial community analysis of mesophilic and thermophilic sludge fermentation processes coupled with thermal hydrolysis, Renew. Sustain. Energy Rev. 141 (2021), 110832, https://doi.org/10.1016/j.rser.2021.110832.
[29] Z.L. Wu, Q. Zhang, Z.Y. Xia, M. Gou, Z.Y. Sun, Y.Q. Tang, The responses of mesophilic and thermophilic anaerobic digestion of municipal sludge to periodic fluctuation disturbance of organic loading rate, Environ. Res. 218 (2023), 114783, https://doi.org/10.1016/j.envres.2022.114783.
[30] B. Eryildiz, M.J. Lukitawesa, Taherzadeh, Effect of pH, substrate loading oxygen and methanogens inhibitors on volatile fatty acid (VFA) production from citrus waste by anaerobic digestion, Bioresour. Technol. 302 (2020), 122800, https://doi.org/10.1016/j.biortech.2020.122800.
[31] J. Yin, X. Yu, Y. Zhang, D. Shen, M. Wang, Y. Long, T. Chen, Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: effect of redox potential and inoculum, Bioresour. Technol. 216 (2016) 996–1003, https://doi.org/10.1016/j.biortech.2016.06.053.
[32] D. Ruan, Z. Zhou, H. Pang, J. Yao, G. Chen, Z. Qiu, Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis, Bioresour. Technol. 289 (2019), 121643, https://doi.org/10.1016/j.biortech.2019.121643.
[33] D. Nguyen, Z. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res. 166 (2019), https://doi.org/10.1016/j.watres.2019.115080.
[34] J.W. Lim, J.A. Chiam, J.Y. Wang, Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste, Bioresour. Technol. 171 (2014) 132–138, https://doi.org/10.1016/j.biortech.2014.08.050.
[35] L.D. Nghiem, P. Manassa, M. Dawson, S.K. Fitzgerald, Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas, Bioresour. Technol. 173 (2014) 443–447, https://doi.org/10.1016/j.biortech.2014.09.052.
[36] C. Vongvichiankul, J. Deebao, W. Khongnakorn, Relationship between pH, oxidation reduction potential (ORP) and biogas production in mesophilic screw anaerobic digester, Energy Procedia 138 (2017) 877–882, https://doi.org/10.1016/j.egypro.2017.10.113.
[37] D. Nguyen, Z. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res. 166 (2019), 115080, https://doi.org/10.1016/j.watres.2019.115080.
[38] Z. Guo, W. Liu, C. Yang, L. Gao, S. Thangavel, L. Wang, Z. He, W. Cai, A. Wang, Computational and experimental analysis of organic degradation positively regulated by bioelectrochemistry in an anaerobic bioreactor system, Water Res. 125 (2017) 170–179, https://doi.org/10.1016/j.watres.2017.08.039.
[39] D.J. Batstone, J. Keller, I. Angelidaki, S.V. Kalyuzhnyi, S.G. Pavlostathis, A. Rozzi, W.T. Sanders, H. Siegrist, V.A. Vavilin, The IWA anaerobic digestion model No 1 (ADM1, Water Sci. Technol. 45 (2002) 65–73, https://doi.org/10.2166/wst.2002.0292.
[40] J. Bai, H. Liu, B. Yin, H. Ma, Modeling of enhanced VFAs production from waste activated sludge by modified ADM1 with improved particle swarm optimization for parameters estimation, Biochem. Eng. J. 103 (2015) 22–31, https://doi.org/10.1016/j.bej.2015.06.015.
[41] D. Botheju, B. Lie, R. Bakke, Oxygen effects in anaerobic digestion, Model. Identif. Control 30 (2009) 191–201, https://doi.org/10.4173/mic.2009.4.1.
[42] A. Sluiter, B. Hames, D. Hyman, C. Payne, R. Ruiz, C. Scarlata, J. Sluiter, D. Templeton, J.W. Nrel, Determination of total solids in biomass and total dissolved solids in liquid process samples, Natl. Renew. Energy Lab. (2008) 3–5.
[43] D.C. Nieves, K. Karimi, I.S. Horváth, Improvement of biogas production from oil palm empty fruit bunches (OPEFB, Ind. Crops Prod. 34 (2011) 1097–1101, https://doi.org/10.1016/j.indcrop.2011.03.022.
[44] APHA-AWWA-WEF, Standard Methods, in: Encycl. Forensic Sci. Second Ed., 23rd ed, Washington, DC, (2017): pp. 522–527. https://doi.org/10.1016/B978–0-12–382165-2.00237–3.
[45] A. Laca, Z. Mousia, M. Díaz, C. Webb, S.S. Pandiella, Distribution of microbial contamination within cereal grains, J. Food Eng. 72 (2006) 332–338, https://doi.org/10.1016/j.jfoodeng.2004.12.012.
[46] E. Simpson, Measurment of diversity, Nature 163 (1949) 688, https://doi.org/10.1038/163688a0.
[47] C.E. Shannon, The mathematical theory of communication, M. D. Comput. 14 (1997) 306–317, https://doi.org/10.2307/410457.
[48] D.J. Batstone, J. Keller, I. Angelidaki, S.V. Kalyuzhnyi, S.G. Pavlostathis, A. Rozzi, W.T. Sanders, H. Siegrist, V.A. Vavilin, Anaerobic Digestion Model No.1 (ADM1), IWA Publishing, (2002).
[49] J. Monod, The growth of bacterial cultures, Annu. Rev. Microb. 3 (1949) 371–394.
[50] L. Lin, R. hong, X. Li, yan Li, Recovery of organic resources from sewage sludge of Al-enhanced primary sedimentation by alkali pretreatment and acidogenic fermentation, J. Clean. Prod. 172 (2018) 3334–3341, https://doi.org/10.1016/j.jclepro.2017.11.199.
[51] X. Huang, C. Duan, J. Yu, W. Dong, H. Wang, Response of VFAs and microbial interspecific interaction to primary sludge fermentation temperature, J. Clean. Prod. 322 (2021), 129081, https://doi.org/10.1016/j.jclepro.2021.129081.
[52] J. Yin, X. Yu, Y. Zhang, D. Shen, M. Wang, Y. Long, T. Chen, Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: effect of redox potential and inoculum, Bioresour. Technol. 216 (2016) 996–1003, https://doi.org/10.1016/j.biortech.2016.06.053.
[53] B.K. Ahring, P. Westermann, Product inhibition of butyrate metabolism by acetate and hydrogen in a thermophilic coculture, Appl. Environ. Microbiol. 54 (1988) 2393–2397, https://doi.org/10.1128/aem.54.10.2393-2397.1988.
[54] S. Xu, A. Selvam, J.W.C. Wong, Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste, Waste Manag. 34 (2014) 363–369, https://doi.org/10.1016/j.wasman.2013.10.038.
[55] L. Zhang, X. Zhou, Q. Gu, M. Liang, S. Mu, B. Zhou, F. Huang, B. Lin, C. Zou, Analysis of the correlation between bacteria and fungi in sugarcane tops silage prior to and after aerobic exposure, Bioresour. Technol. 291 (2019), 121835, https://doi.org/10.1016/j.biortech.2019.121835.
[56] X. Zhou, Y. Lu, L. Huang, Q. Zhang, X. Wang, J. Zhu, Effect of pH on volatile fatty acid production and the microbial community during anaerobic digestion of Chinese cabbage waste, Bioresour. Technol. 336 (2021), https://doi.org/10.1016/j.biortech.2021.125338.
[57] H. Ren, Y. Feng, T. Liu, J. Li, Z. Wang, S. Fu, Y. Zheng, Z. Peng, Effects of different simulated seasonal temperatures on the fermentation characteristics and microbial community diversities of the maize straw and cabbage waste co-ensiling system, Sci. Total Environ. 708 (2020), 135113, https://doi.org/10.1016/j.scitotenv.2019.135113.
[58] J. Lee, J.R. Kim, S. Jeong, J. Cho, J.Y. Kim, Long-term performance of anaerobic digestion for crop residues containing heavy metals and response of microbial communities, Waste Manag. 59 (2017) 498–507, https://doi.org/10.1016/j.wasman.2016.10.005.
[59] L. Song, Y. Song, D. Li, R. Liu, Q. Niu, The auto fluorescence characteristics specific activity and microbial community structure in batch tests of mono-chicken manure digestion, Waste Manag. 83 (2019) 57–67, https://doi.org/10.1016/j.wasman.2018.11.002.
[60] D. Poggio, M. Walker, W. Nimmo, L. Ma, M. Pourkashanian, Modelling the anaerobic digestion of solid organic waste – substrate characterisation method for ADM1 using a combined biochemical and kinetic parameter estimation approach 53 (2016) 40–54, https://doi.org/10.1016/j.wasman.2016.04.024.
[61] H. Zhou, S. Lv, Z. Ying, Y. Wang, J. Liu, W. Liu, Characteristics of two-phase mesophilic anaerobic digestion of co-substrates consisting of waste activated sludge and corn silage based on modified ADM1, Waste Manag. 91 (2019) 168–178, https://doi.org/10.1016/j.wasman.2019.05.006.